

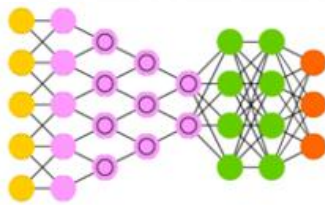
DEEP ARCHITECTURES

NEURAL NETWORKS

DROPOUT

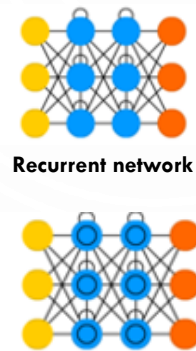
- Deactivate n neurons.
 - In the Keras library, a percentage is specified.
- Helps to mitigate the *vanishing gradient* problem.

CLASSIC DEEP ARCHITECTURES

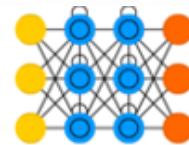


**Convolutional
neural network**

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Recurrent network



LSTM network



Autoencoder



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WHAT ARE THEY USED FOR?

- Some of these architectures are quite useful for data with dependencies.

Architecture	Use
Convolutional neural network	Spatial dependencies
LSTM network	Temporal dependencies
Autoencoder	Compression

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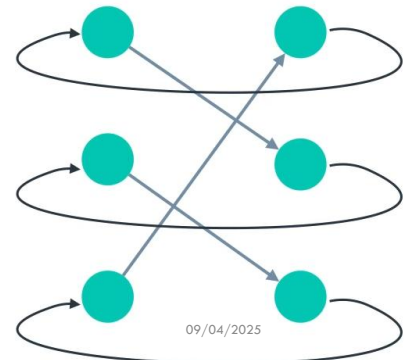
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RECURRENT NEURAL NETWORKS

- Hidden states for remembering temporal data
- Outputs are used as inputs

<https://www.youtube.com/watch?v=UNmqTiOnRfg>



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RECURRENT NEURAL NETWORKS

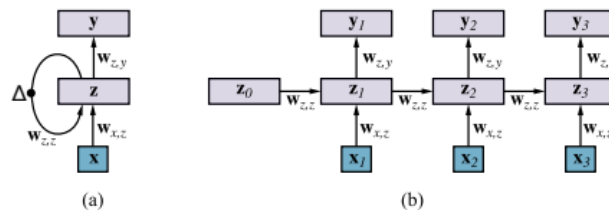


Figure 21.8 (a) Schematic diagram of a basic RNN where the hidden layer z has recurrent connections; the Δ symbol indicates a delay. (b) The same network unrolled over three time steps to create a feedforward network. Note that the weights are shared across all time steps.

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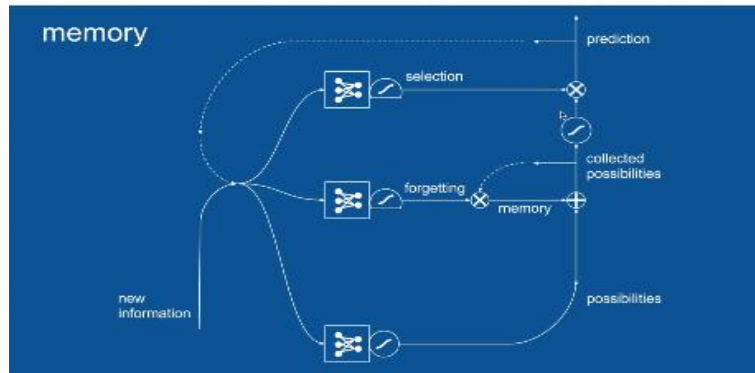
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LONG-SHORT TERM MEMORY (LSTM) NETWORKS

- Type of recurrent NN
- “Forgetting” gates

<https://www.youtube.com/watch?v=WCUNPb-5EYI>

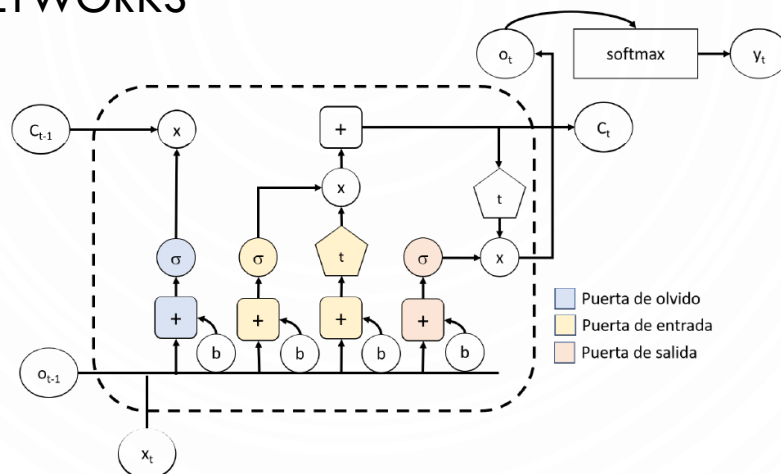


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LSTM NETWORKS



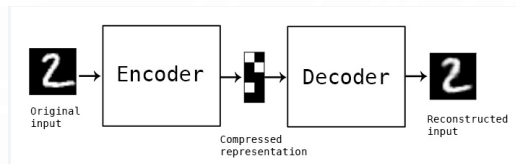
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AUTOENCODERS

- Have a “narrow” layer to summarize or compress data using a few neurons.
- Decoding layers may also be included.



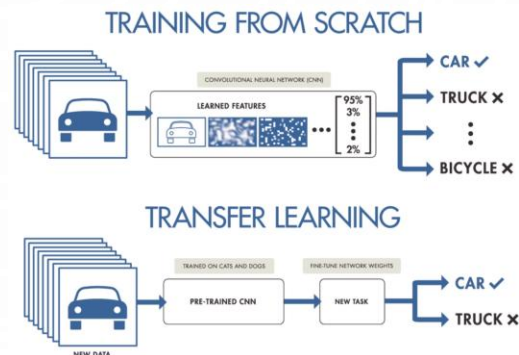
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TRANSFER LEARNING

- Use of pre-trained models.
 - Example: MobileNet V2
- Part of the model is kept (“frozen layers”), but the last layer(s) is changed.
- Can be applied when having similar tasks.



Source:

https://miro.medium.com/v2/resize:fit:1000/0*xNIEPIZmPvKegss6

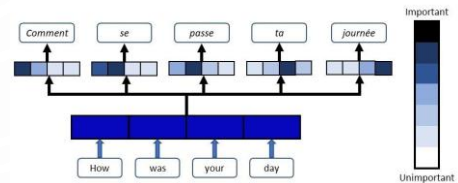
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ATTENTION MODELS

- The network is focused on specific aspects. Example: particular words of a text
- Task decomposition
- Input processing mechanism



Source:

<https://blog.floydhub.com/content/images/2019/09/Slide36.JPG>

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TRANSFORMERS

- Used for natural language processing (NLP) tasks
- Based on
 - Attention mechanisms (*multi-head attention*)
 - Positional encoding

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TRANSFORMER EXAMPLES

- DALL-E: <https://openai.com/dall-e-2/#demos>
- ChatGPT: <https://chat.openai.com/>



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APPLICATIONS

- Computer vision
- Automatic translation
- Predictive text
- Content generation (GPT-3, GPT-4)
- Autonomous vehicles
- Chatbots and conversational agents



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REFERENCES

- Deep AI. Attention Models. <https://deepai.org/machine-learning-glossary-and-terms/attention-models>